Vikramaditya Tatke

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Data Engineer and team leader with four years of experience driving impactful data solutions and database administration. Proficient in developing scalable data pipelines, integrating REST APIs, and optimizing analytical workloads across diverse sectors. Expertise includes a range of technologies, from Python-based frameworks to cloud platforms and database systems. Known for a proactive approach to problem-solving and enthusiasm for open source.

SKILLS

- **Programming Languages:** Python (FastAPI, Django, Pydantic, Boto3, REST APIs), SQL, Java, Rust.
- **Data Engineering Frameworks:** Apache Kafka, Apache Airflow, Dagster, Pandas/Polars, PySpark.
- **Databases:** ClickHouse, MongoDB, Postgres, DuckDB, Redis.
- **Cloud:** AWS EC2, AWS Lambda, AWS Redshift, AWS Event Bridge Scheduler, AWS Secrets Manager, MS Azure Sentinel.
- **DevOps and Version Control:** Docker, Jira, Git (Bitbucket and GitHub), GitHub Actions.
- **Business Intelligence and Data Visualization:** Tableau, Qlik Sense, Evidence, Streamlit, Matplotlib.

CERTIFICATIONS

ClickHouse Certified Developer: *Issued March 2025.*

MongoDB Associate Developer: *Issued February 2024.*

Tableau Desktop Specialist: *Issued July 2020.*

WORK EXPERIENCE

SecurityHQ — **London, United Kingdom**

Lead Data Engineer — *December 2024–Present*

- Mentored and guided a team of 5 platform engineers, providing technical expertise and project oversight for ETL and database development, acting as the senior code reviewer.
- Proposed and implemented internal improvement initiatives for the data platform, focusing on POCs to evaluate new technologies and enhance data processing capabilities.
- Proactively optimized analytical workloads and database performance by analysing query execution plans, identifying bottlenecks, and implementing indexing strategies (aligns with advanced SQL proficiency).
- Executed a seamless migration from self-hosted MongoDB to ClickHouse Cloud using AWS DMS, improving scalability and performance, ultimately cutting annual costs by 75% and boosting analytical query performance by over 1000 times.
- Created CI/CD workflows for deploying DAGs to Airflow using GitHub Actions.
- Designed and implemented data pipelines using Apache Airflow and created CI/CD workflows for deploying these DAGs using GitHub Actions.

Data Engineer — *June 2022–December 2024*

- Architected and developed a scalable, high-throughput, Python-based data pipeline, leveraging asynchronous programming to reliably stream over 10TB of event logs weekly.
- Designed and built robust data pipelines to ingest diverse data formats (JSON, CSV, Parquet) from REST APIs and AWS S3, demonstrating proficiency in API integration using Apache Airflow.
- Improved database memory management by utilizing best practices for data storage and compression and session management.
- Collaborated with stakeholders to translate business requirements into robust data pipeline designs, bridging technical needs with strategic objectives.
- Ensured high standards of data quality and accuracy throughout the data lifecycle for clients across diverse sectors including financial services, construction, government, and insurance.

Advanced Manufacturing Control Systems (AMCS) Group **Limerick, Ireland**
Associate Software Support Engineer *February 2022–June 2022*

- Optimized SQL Server databases for customers ensuring data integrity and query performance using best practices.
- Monitored CPU utilization, memory usage, and disk I/O in customer environments to identify potential issues proactively.
- Diagnosed and resolved complex database issues, including system errors, performance bottlenecks, and compatibility problems.

Amazon Web Services **Dublin, Ireland**
Software Development Engineer 1 Intern *May 2021–November 2021*

- Developed a critical component for the DynamoDB meta-data team that promised 99.999999999% availability.
- Migrated client discovery service from Java to Rust to overcome the drawbacks of the garbage collector and target consistent sub-1ms client-side latency.
- Optimized the service by implementing multi-threading, an asynchronous network runtime and data compression over the wire.

PROJECTS

Real-time Event Processing Pipeline with Flink and Kafka

- Designed and implemented a distributed data pipeline for real-time Windows event log processing using Java, Apache Flink, and Kafka. Demonstrated fault-tolerant stream processing with TCP ingestion, JSON parsing, and Kafka integration.

Polars GPU Acceleration Benchmark

- Implemented and benchmarked a high-performance data processing pipeline utilizing Polars with GPU acceleration. Developed comprehensive Jupyter Notebooks to analyse and visualize performance metrics on large datasets.

Streamlit Admin Panel

- Designed and implemented a fully functional admin panel using Streamlit for real-time data visualization and management. Developed Python scripts to handle backend data processing and integration with various data sources. Ensured seamless user interaction and efficient data handling within the application.

QRadar REST API Kafka Data Pipeline

- Developed a data pipeline to integrate QRadar REST API with Kafka for real-time data ingestion and processing. Utilized Python to implement API calls, data transformation, and Kafka producer/consumer logic. Implemented well-structured logging using Python's logging module, ensuring detailed logs are stored efficiently for monitoring and troubleshooting. Logs are stored in a centralized logging system for easy access and analysis.

EDUCATION

National College of Ireland **Dublin, Ireland**
Post Graduate Diploma in Science in Data Analytics *September 2018–March 2020*
MITSOM College, Pune University **Pune, India**
Bachelors in computer applications (BCA) *June 2014–May 2017*